SME Credit Scorecard Development

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Background & Data Overview

SME portfolio scorecard — expert-weighted system under review

Context

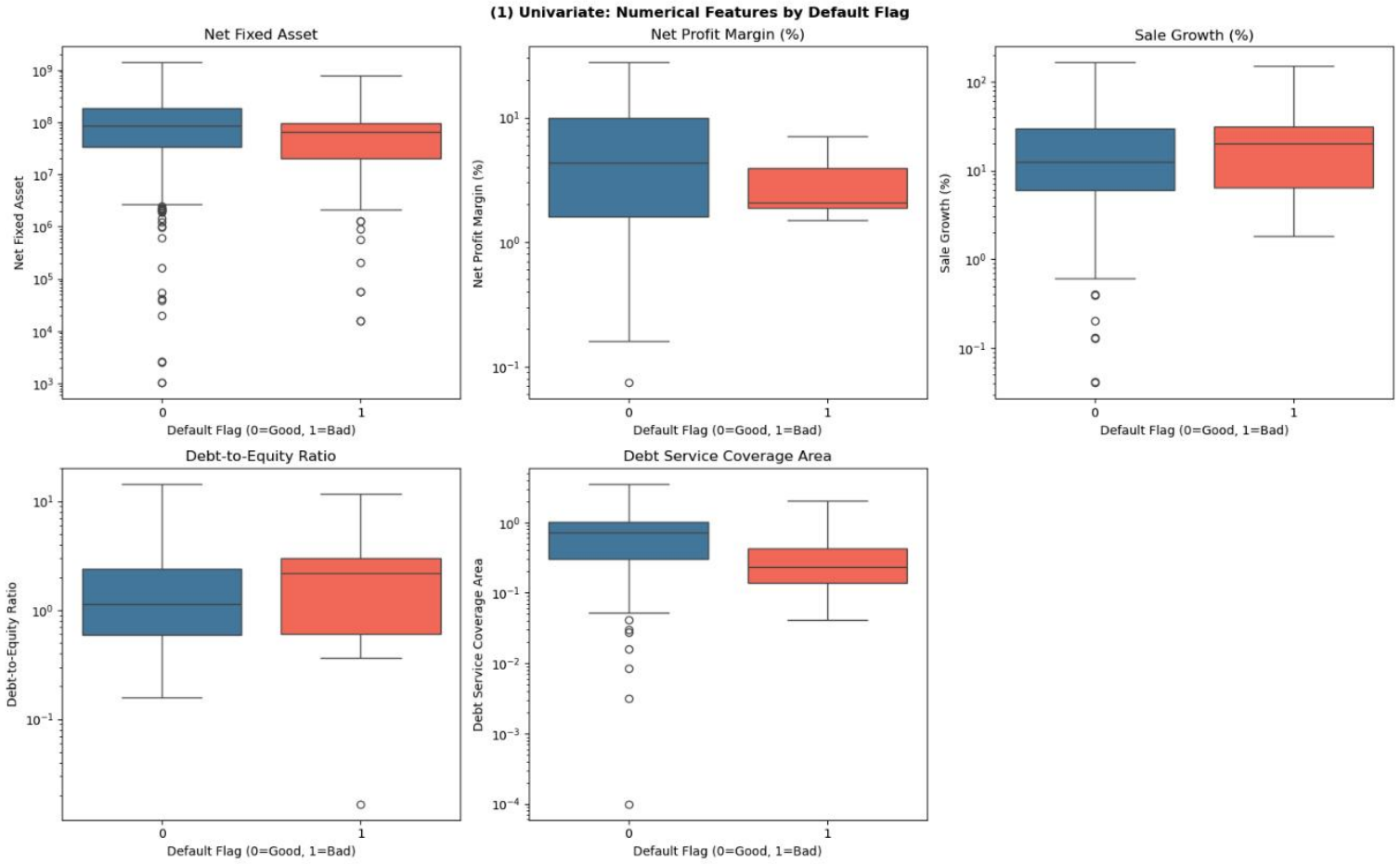
Dataset	557 SME borrowers, 8 features, 9.0% default rate
Target	Default Flag (1 = default, 0 = non default)
Scorecard	Expert-weighted, C + F components, score per bin
C Component	50% weight
F Component	50% weight: F1–F5 financial ratios equally at 10% each
Missing Data	C2: 282 NaN (51%), Net Fixed Asset: 26 (5%)
Sentinels	99999998/99999999
Objective	Assess expert weights using quantitative analysis ; develop LR-based PD model

Feature Summary

Code	Feature	Type	IV	Expert Wt
C1	Debt Restructuring History	Cat	1.17	12.5%
C2	Debt Rescheduling History	Cat	0.57	12.5%
C3	Worst CB Delinquency 2yr	Cat	5.43	25.0%
F1	Net Fixed Asset	Num	2.60	10.0%
F2	Net Profit Margin (%)	Num	1.28	10.0%
F3	Sale Growth (%)	Num	0.04	10.0%
F4	Debt-to-Equity Ratio	Num	1.46	10.0%
F5	Debt Service Coverage Ratio	Num	1.58	10.0%

EDA: Univariate Analysis

Distribution and default rates across categorical and numerical features



Key Observations

- **F1 - Net Fix Asset**

Data have high variance and outliers. Fix asset of firms that not defaults have higher mean

- **F4 - DSCR**

Defaulters concentrated at lower DSCR; non-defaulters spread across positive values

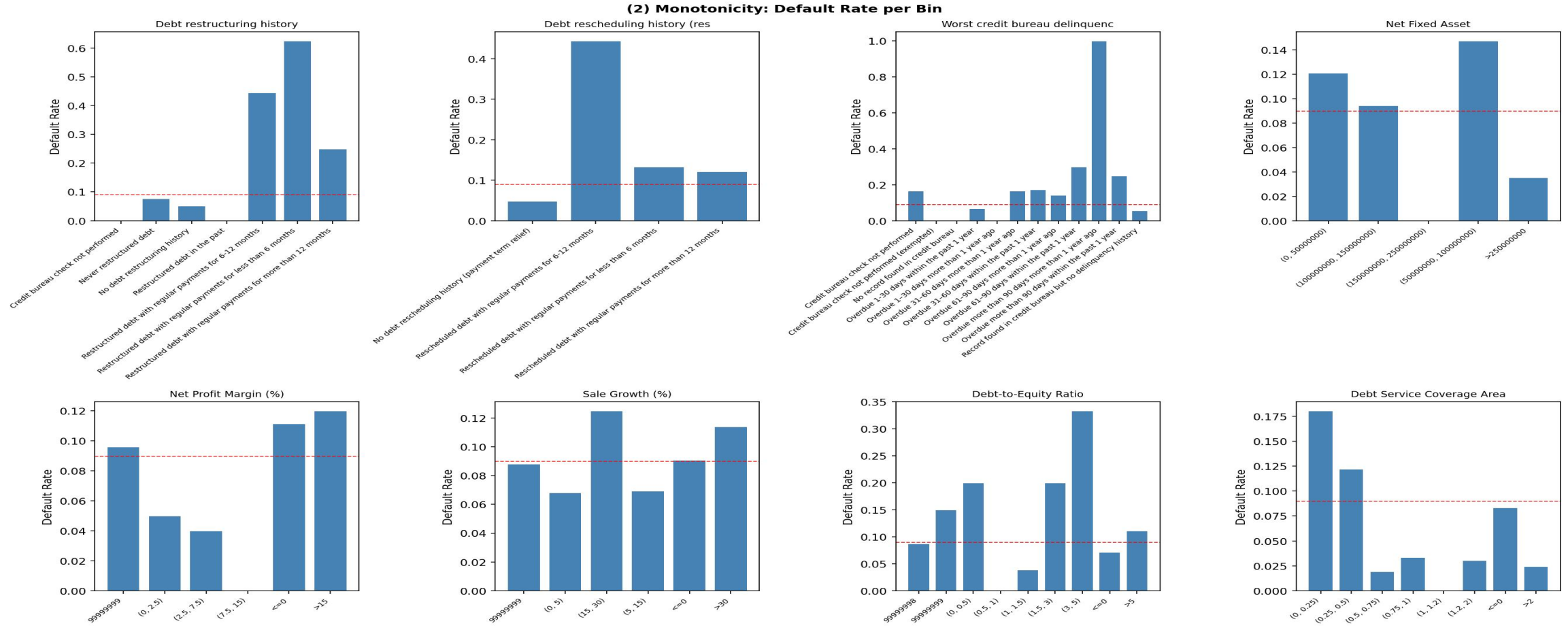
- **F2 - Net Profit Margin**

Heavy left-skew, median near -8%; most SMEs unprofitable

- **F3 - Sale Growth**

Quantile 1 and 3 of defaulter and non defaulter are approximately the same, many zero-growth observations

EDA: Monotonicity Analysis



Monotonicity Findings:

- **C1 Restructuring:**

Non-monotone: "Never restructured" has lower WoE than "No history" - bins should be merged

- **C3 CB Delinquency:**

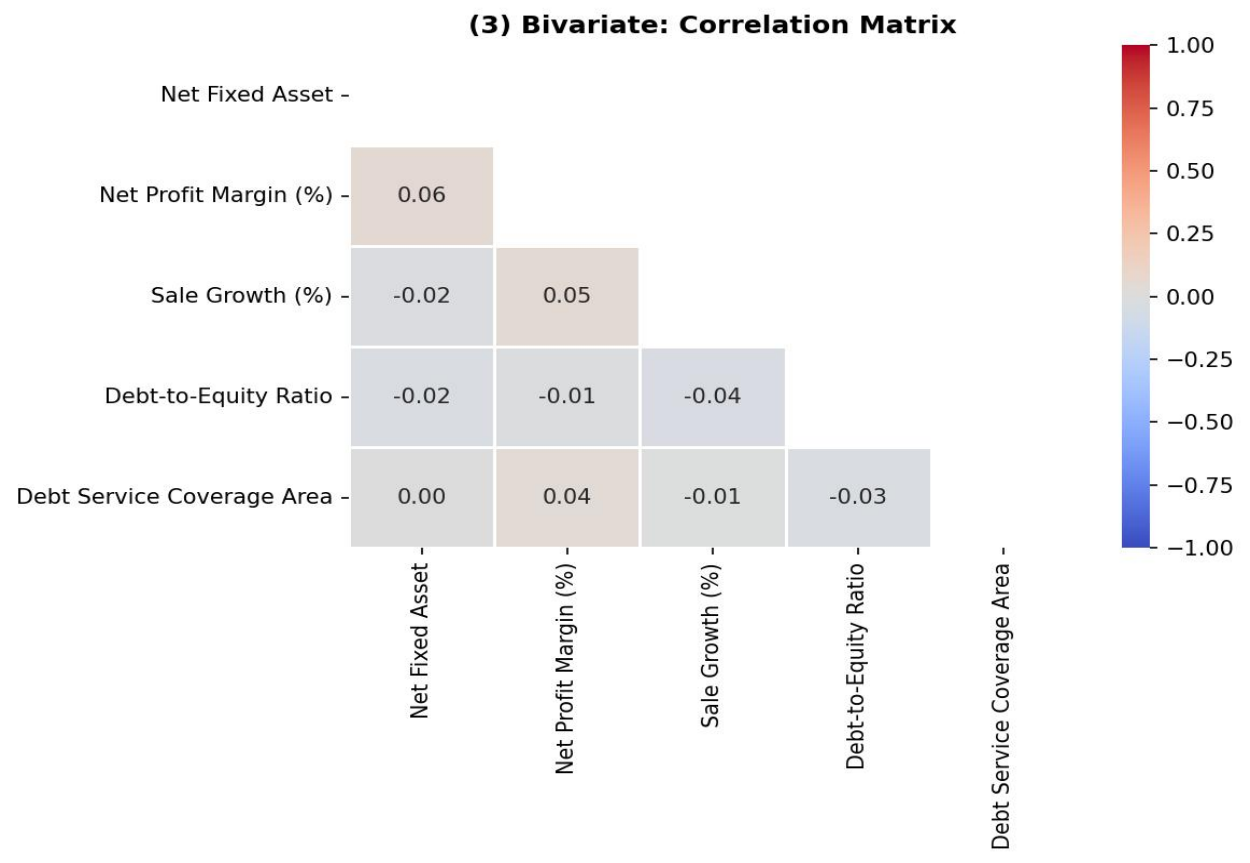
Broadly monotone: longer overdue → higher WoE

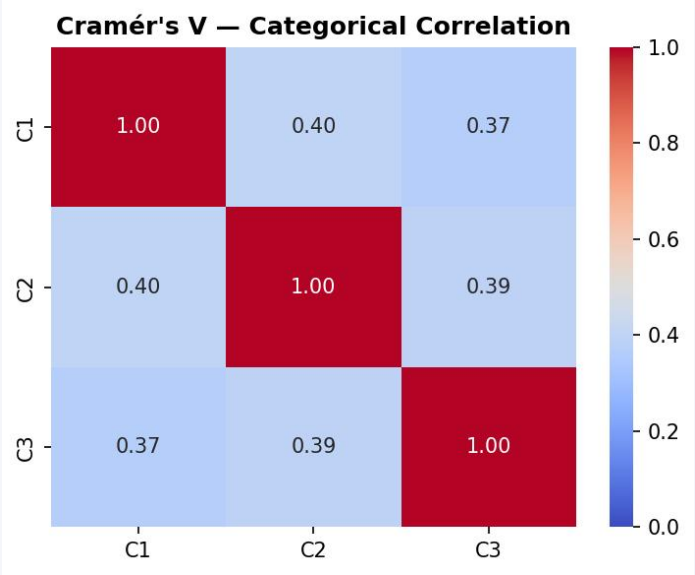
- **F4 D/E Ratio:**

Partially monotone; (0.5,1) bin has zero defaults

EDA: Correlation & Multicollinearity

Pearson correlation for numerical features; Cramér's V for categorical pairs





Multicollinearity Findings

- Cramér's V: moderate correlation among credit history features
- Net Profit Margin and Net FIx Asset show weak correlation
- No severe multicollinearity

Q1: Feature Weight Assessment

Apprach 1: Information Value (IV) as objective measure vs expert-assigned weights

Feature Weight Comparison: Expert vs IV-based

Feature	Expert Weight	IV	Ranking
C1_Debt restructuring history	12.5%	1.17	Strong
C2_Debt rescheduling history (reschedule)	12.5%	0.57	Strong
C3_Worst credit bureau delinquency record in the past 2 years	25.0%	5.43	Strong
F1_Net Fixed Asset	10.0%	2.60	Strong
F2_Net Profit Margin (%)	10.0%	1.28	Strong
F3_Sale Growth (%)	10.0%	0.04	Weak
F4_Debt-to-Equity Ratio	10.0%	1.46	Strong
F5_Debt Service Coverage Area	10.0%	1.58	Strong

(*) IV calculation and ranking from book Intelligent credit scoring - Naeem Siddiqi

Recommendation: Proposed Weight Rebalancing

Using IV to assess predictive power:
- Sale Growth (IV=0.04) shows negligible signal and **should be removed**.
- F1_Net Fixed Asset outperform F2_Net Profit Margin and F4_Dept-to-Equity Ratio
- C1_Dept restructuring history have more predictive power than C2_Dept rescheduling history (reschedule)

Q1: Feature Weight Assessment

Approach 2: Calculating weights using linear regression model

LR Coefficients & Normalised Importance

Feature	Coef	LR Wt%	Expert
C3 Worst CB Delinquency	0.897	26.60%	25%
F4 Debt-to-Equity Ratio	0.745	22.17%	10%
F5 DSCR	0.545	16.22%	10%
C2 Debt Rescheduling	0.469	13.96%	12.5%
F1 Net Fixed Asset	0.282	8.43%	10%
C1 Debt Restructuring	0.238	7.09%	12.5%
F2 Net Profit Margin	0.182	5.52%	10%

Recommendation: Proposed Weight Rebalancing

Increase weights on highest-impact leverage
(1) Debt-to-Equity Ratio - second strongest predictive feature (currently underweighted by ~12 %)
(2) Debt Service Coverage Ratio (DSCR) - underweighted by ~ 6%
Decrease weights on features
(1) Debt Restructuring History
(2) Net Profit Margin

Q2: Scorecard Enhancement Suggestions

Approach : Validate monotonicity - default rate must strictly increase band-by-band

Feature	Sale Growth (%)

Bin	Default Count.	Total Count.	Default Rate %	Expert Scorecard score
(0, 5)	3	44	6.82	25
(15, 30)	6	48	12.5	75
(5, 15)	5	72	6.94	50
<=0	21	231	9.09	0
>30	9	79	11.39	100
Not available	6	83	7.23	0
Grand Total	**50**	**557**	**53.97**	**250**

Recommendation: F3 Sale Growths - cap binning the score

Current scorecard score contradict default rate trend
Higher sales growth is associated with greater default risk
> Suggesting that aggressive growth may signal financial distress rather than stability.

Q3: EDA Methodology for Credit Risk Datasets

Structured five-step framework for borrower, financial and macroeconomic data

Step 1 Data Quality	Step 2 Univariate	Step 3 Bivariate	Step 4 IV/WoE	Step 5 Macroeconomic
• Missing value audit & sentinel detection • Skewness check, IQR-based outlier treatment • Duplicate and data type validation	• Categorical: frequency, default rate per bin, concentration check • Numerical: histogram, boxplot, describe() • Flag features with >50% single-bin concentration	• Numerical vs Target: KS test, Pearson correlation • Categorical vs Target: Chi-square, Cramér's V	• Quantify predictive power per bin	• Usually affect the whole portforlio or segment • Shoud be consider as a adverse or favorable variable • Should be monitor in at least 1 business cycle • Two-stage model: micro scorecard + macro adjustment

Q4: PD Model Development — Logistic Regression

Theory, assumptions and implementation steps

Model Theory

- **General formulation**

WoE encoding linearises the log-odds relationship

$$f(x) = \theta(w^T x)$$

- **MLE estimation**

Parameters estimated by minimizing negative log-likelihood function

$$J(w) = -\sum_{i=1}^{N} (y_i \log z_i + (1 - y_i) \log(1 - z_i))$$

- **Have no close form soluion**

Approximate close solution by iterating $w = w + \eta(y_i - z_i)x_i$

Assumption

- **Binary or Ordinal Dependent Variable**
- **Independence of Observations**
- **Absence of Multicollinearity**
- **Linearity of Independent Variables with the Log-Odds**
- **No extreme outliers**

Implementation Steps

1 **1. Data Preparation**

2 **2. WoE Encoding**

Compute WoE per bin

3 **3. Train/Test Split**

80/20 stratified split; EPV check (need ≥10 defaults per feature)

4 **4. Model Training**

Set stopping condition when loss funcion small enough or by iteration

5 **5. Validation**

6 **5. Score Mapping**

$\text{Score_i} = \text{Offset} + \text{Factor} \times \beta_0 + \Sigma[-\text{Factor} \times \beta_i \times \text{WoE}_i]$

7 **6. PDO Calibration**

Allocating rating classes to entities to ensure the calibration power of the division created is as high as possible

Q5: Model Evaluation Framework

Three pillars: goodness-of-fit, discriminatory power, accuracy

Pillar 1
Goodness-of-Fit

Hosmer-Lemeshow Test

Groups: 10
Hosmer-Lemeshow Statistic: 4.8454
Degrees of Freedom: 8
P-value: 0.7740
Good fit — fail to reject H0

Binomial Test

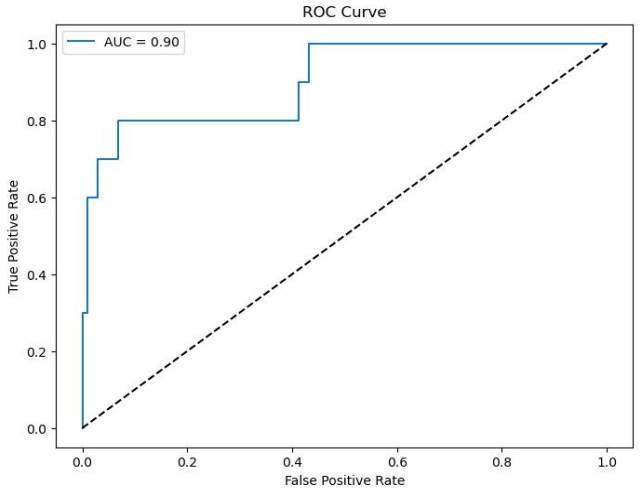
Observed Defaults: 10
Total Borrowers: 112
Expected PD (avg): 0.116
Expected Defaults: 12
p-value: 0.46

Fail to reject H0 — predicted default rate matches observed

Pillar 2
Discriminatory Power

AUC-ROC = 0.90

Model ranks 90% of default/non-default pairs correctly
Excellent - robust even with extreme WoE values
Gini Coefficient = 2×AUC−1 = 0.80



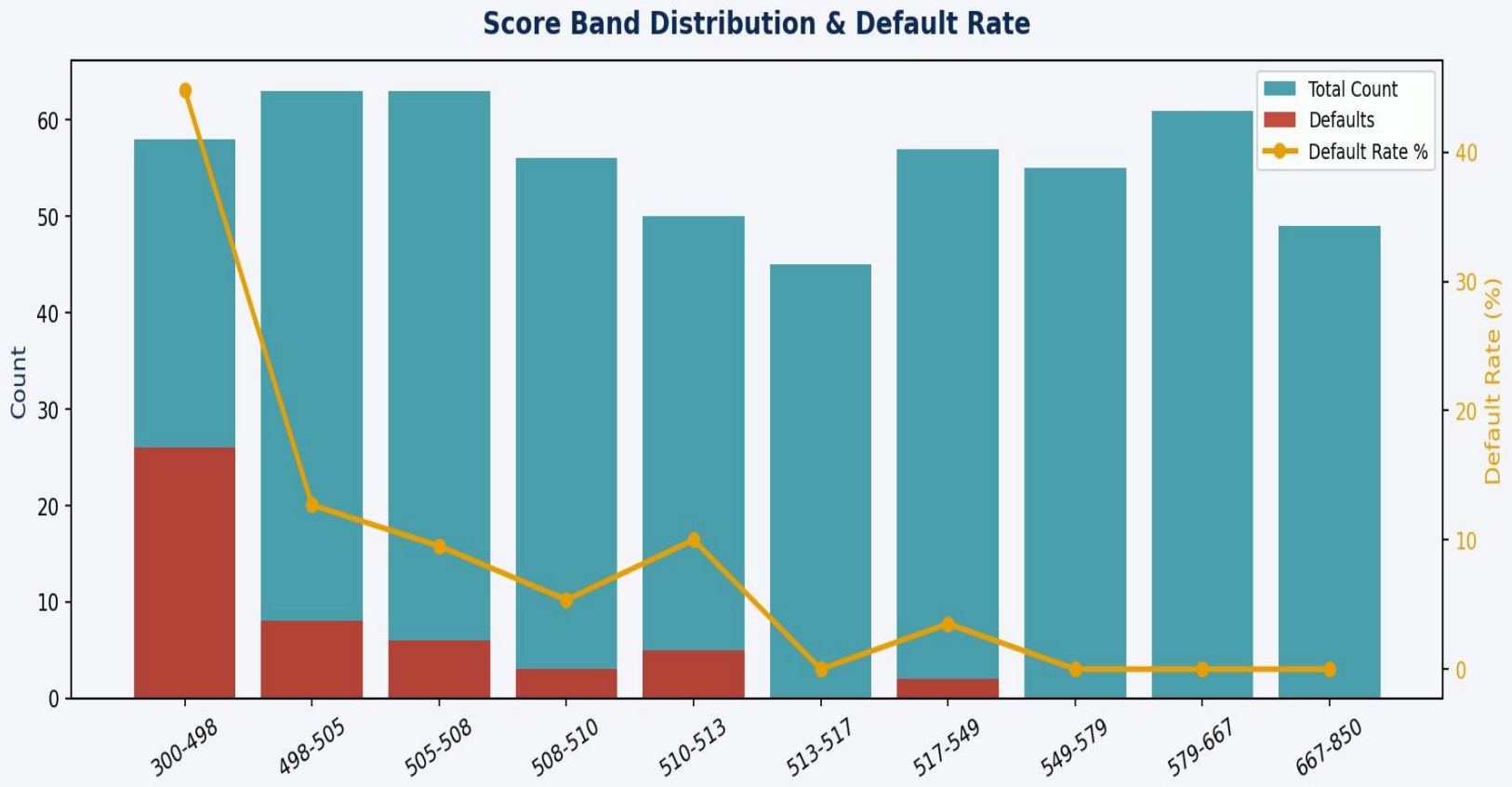
Pillar 3
Forecasting Accuracy

PSI — Population Stability

	precision	recall	f1-score	support
0	0.96	0.99	0.98	102
1	0.86	0.60	0.71	10
accuracy			0.96	112
macro avg	0.91	0.80	0.84	112
weighted avg	0.95	0.96	0.95	112

Score Distribution Analysis

PDO-calibrated scores mapped to 300–850 range with band-level default rates



PDO Calibration

PDO	20 (score doubles odds per 20 pts)
Base Score	600 at odds 50:1
Factor	PDO/ln(2) = 28.85
Offset	Base – Factor×ln(Base_Odds) = 487.1
Intercept	Factor × β_0 = 10.55
Score Range	300–850 (min-max scaled)

Score Band Findings

- 300–498: 44.8% default rate → High Risk
- 498–513: 5–13% default rate → Medium Risk
- 513+: 0–3.5% default rate → Low Risk
- Clear separation validates model discrimination
- Bands 513–850: 0 defaults in top 5 bands

Conclusion

Based on quantitative analysis, three actions are recommended to strengthen the scorecard:

01 Reassess the Weights of Risk Components

- C3 (Worst CB Delinquency) is the strongest predictor with IV = 5.43 — its 25% weight is justified and may warrant an increase to 30%
- F3 Sale Growth has IV = 0.04, below the acceptable threshold — it should be removed or replaced with a more predictive variable
- D/E Ratio and DSCR are underweighted relative to their data-implied importance — consider increasing both to 15%
- Use IV as the primary basis for weight assignment rather than expert judgement alone

02 Adjust Bins and the Formulated Scorecard

- Merge non-monotone bins where WoE direction breaks the expected risk ordering (e.g. C1 Restructuring)
- Cap extreme values at 5th–95th percentile before binning to reduce sensitivity to outliers in skewed features
- Validate monotonicity after each bin adjustment — default rate must strictly increase from best to worst bin

03 Combine Quantitative Model with Expert Model

- Use LR-implied coefficients to cross-validate expert weights — flag features where the gap exceeds 5 percentage points
- Adopt a hybrid approach: data-driven weights as the baseline, with expert overlay for features with limited historical data
- Incorporate macroeconomic variables (GDP growth, interest rates) in a two-stage model: micro scorecard × macro scalar